AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII EXAMINATION

APRIL/ MAY 2017

Chemistry Paper II

Time: 2 hours 20 minutes Marks: 55

AKU-EB May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are ELEVEN questions. Answer ALL questions. Questions 10 & 11 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 5 Marks)

a. Identify the period, group and block of an element based on its electronic configuration given below. (3 Marks)

$$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^{10}, 4p^6$$

Period: _____

Group: _____

Block: _____

b. Write the decomposition products of the given nitrate of magnesium (Mg). (2 Marks)

$$2Mg(NO_3)_{2(s)} \longrightarrow _____ + _____ + O_{2(g)}$$

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Q.2. (Total 5 Marks)

a. Give suitable reasons why

i. transition elements show variable oxidation states. (1 Mark)

ii. different transition metal ions produce different colours in complex compounds. (1 Mark)

iii. transition elements are used as catalyst. (1 Mark)

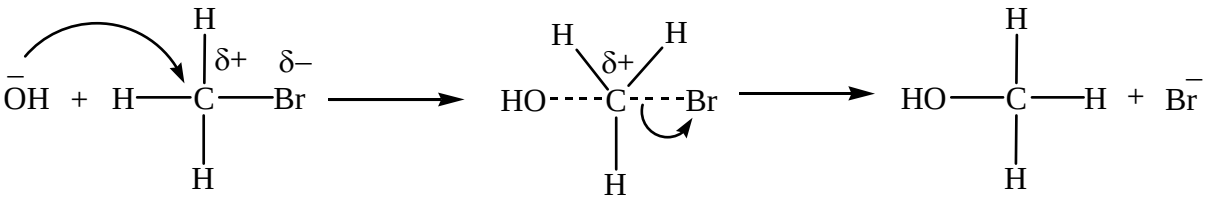
b. Identify the coordination number of Cu and Fe in the given coordination compounds.

i. $[Cu(NH_3)_4]SO_4$ (1 Mark)

ii. $K_4[Fe(CN)_6]$ (1 Mark)

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Q.3. (Total 4 Marks)



Transition state

Identify the following in the given mechanism of nucleophilic substitution (SN) reaction.

a. Type of nucleophilic substitution reaction (1 Mark)

b. Type of alkyl halide (R–X) used (1 Mark)

c. Leaving group (1 Mark)

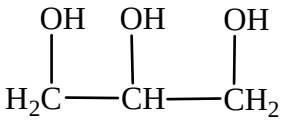
d. Number of molecules involved in the rate determining step (1 Mark)

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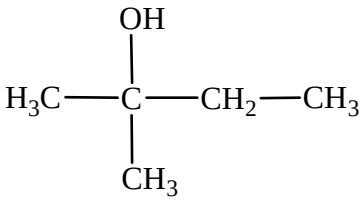
Q.4. (Total 5 Marks)

a. Write the IUPAC names of the following alcohols.

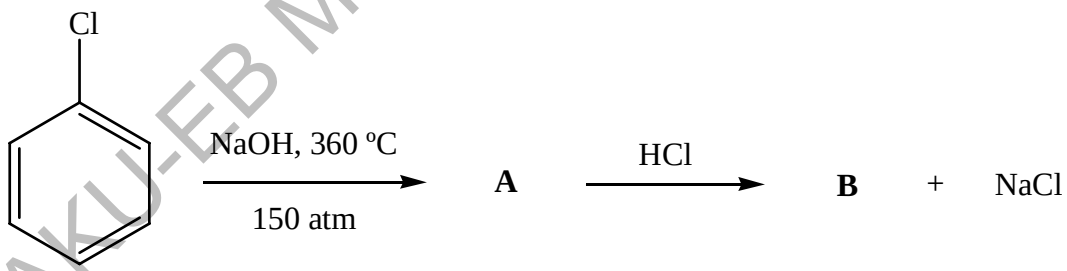
i. (1 Mark)



ii. (1 Mark)



b. Complete the given chemical reaction by drawing the structures of products **A** and **B**. (2 Marks)



Product **A**	Product **B**

c. Why are ethers slightly soluble in water? (1 Mark)

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Q.5. (Total 4 Marks)

Write the mechanism for the given reactions of ethene.

a. $H_2C{=}CH_2 + O_3 \longrightarrow 2\ H{-}C(=O){-}H + H_2O_2$ (2 Marks)

Space for mechanism

b. $H_2C{=}CH_2 + Br_2 \longrightarrow H_2C(Br){-}CH_2(Br)$ (2 Marks)

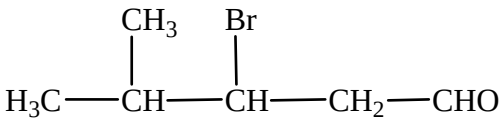
Space for mechanism

B May Ex

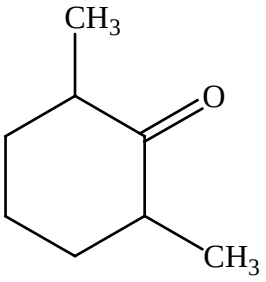
Q.6. (Total 4 Marks)

a. Write the IUPAC names of the following carbonyl compounds.

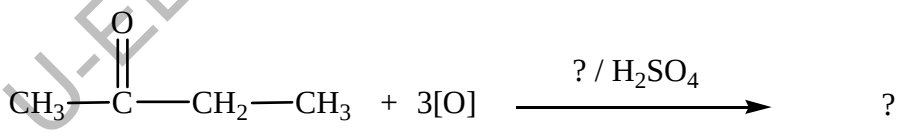
i. (1 Mark)



ii. (1 Mark)



b. Identify the oxidising agent and the product(s) in the given chemical reaction. (2 Marks)



Oxidising agent: _____

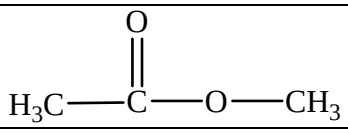
Product(s): _____

_____

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Q.7. (Total 4 Marks)

Identify the functional group present in each of the given molecules**.**

S. No.	Molecule	Functional Group
i	$CH_3—CH_2—NH_2$	
ii	$H_3C—H_2C—O—CH_2—CH_3$	
iii		
iv	$CH_3—CH_2—SH$	

Q.8. (Total 5 Marks)

a. What is greenhouse effect? (1 Mark)

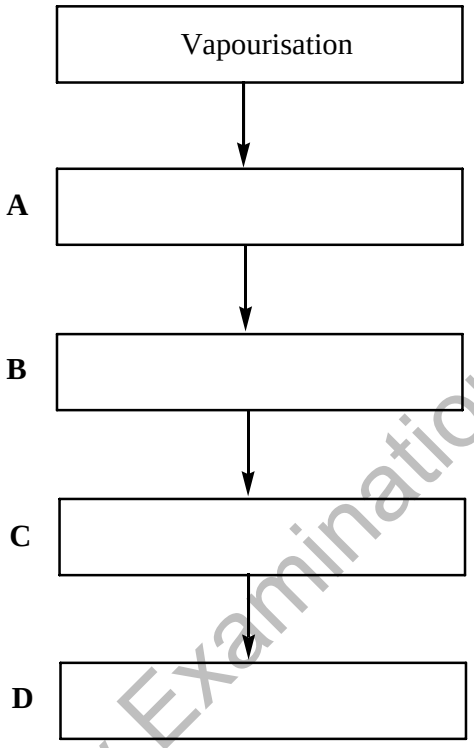
b. Why is greenhouse effect important for normal life on Earth? (1 Mark)

c. Write any THREE impacts of increased concentration of carbon dioxide on the environment. (3 Marks)

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Q.9. (Total 4 Marks)

Name the four stages, **A** to **D,** of a mass spectrometer in sequence in the given boxes.



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Q.10. (Total 8 Marks)

EITHER

a. Describe the addition reactions of ethene with the following electrophilic reagents using balanced chemical equations. Mention conditions, i.e. catalyst, temperature and/ or pressure, where essential.

i. Sulphuric acid (hydration) (2 Marks)
ii. Hydrogen bromide (2 Marks)
iii. Hypochlorous acid (2 Marks)
iv. Hydrogen (hydrogenation) (2 Marks)

OR

b.

i. Write the steps of mechanism for the following reaction. (3 Marks)



ii. Write chemical equations to show reaction of carboxylic acid (CH_3COOH) with the given reagents. Mention catalyst where required. (5 Marks)

I. C_2H_5OH
II. CH_3CH_2COOH
III. PCl_5

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Q.11. (Total 7 Marks)

EITHER

a.

i. Write the role of each of the given pesticides. (4 Marks)

I. Molluscicides
II. Rodenticides
III. Herbicides
IV. Fungicides

ii. What are dyes? What is the colour of methyl orange in (3 Marks)

I. acidic medium?
II. basic medium?

OR

b.

i. What are proteins? (1 Mark)

ii. Describe simple, conjugated and derived proteins. Give ONE example of each type. (6 Marks)

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